

A weighted determinant polynomial with satellites

MathlibAnnex.Satellite.satellitePolynomial

def

Adds linear row-replacement contributions to a base determinant.

Statement

Let $n \in \mathbb{N}$, $V = \mathbb{R}^n$ with reference sup norm, and J a finite index set. Let $F = (\ell_i)_{i=1}^n$ be a family of continuous real linear functionals on V , and $s = (s_a)_{a \in J}$ another such family. Choose $w \in \mathbb{R}$ and coefficients $c_a \in V$. Let $e_1, \dots, e_n$ be the standard coordinate vectors. Put $M_F = (\ell_i(e_j))_{i,j}$ and replace row i by $(s_a(e_j))_j$ when writing $M_F[i \leftarrow s_a]$.

Definition

The full defining expression is

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a].$$

Both sums are finite. If J is empty, only $w \det M_F$ remains. If $n = 0$, the empty determinant is 1 and the inner sum is empty, so the value is w .

Assumptions

Only the finite indices, the real weight, coefficient vectors and the two families of continuous linear functionals are inputs. There is no norm model p , contraction condition, positive-weight hypothesis or invertibility hypothesis.

Conclusion

The definition returns the scalar $\Phi_{w,c}(F, s)$. The satellites are additional dual functionals, not vectors or points of a net.

The private coordinate and row-replacement aliases in the exact local aliases identify reference vectors, frame rows, and standard-basis matrices. They do not restrict the defining polynomial to admissible configurations; such a restriction is imposed only in later theorems.

Main citations

- the exact local aliases — Satellite local aliases
- Exact declaration and proof — MathlibAnnex.Satellite.satellitePolynomial

Lean source signature (exact)

```
def satellitePolynomial {n : ℕ} {J : Type*} [Fintype J]
  (weight : ℝ) (coeff : J → Coord n)
  (B : DualFrame n) (sat : J → (Coord n →L[ℝ] ℝ)) : ℝ :=
  weight * frameDet B +
  ∑ a : J, ∑ i : Fin n, coeff a i * replacementDet B i (sat a)
```

This is a definition. All displayed arguments are given; := introduces its complete right-hand side.

In the source	Mathematical meaning
<code>n; {J : Type*} [Fintype J]</code>	A nonnegative integer n and a finite index set J , possibly empty.
<code>Coord n</code>	The reference space $V = \mathbb{R}^n$ with its sup norm; <code>Coord n</code> abbreviates <code>Fin n → ℝ</code> . Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
<code>(weight : ℝ); (coeff : J → Coord n)</code>	The real weight w and the coefficient family $a \mapsto c_a \in V$.
<code>(B : DualFrame n)</code>	The base family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals on V .
<code>(sat : J → (Coord n →L[ℝ] ℝ))</code>	The additional family $a \mapsto s_a$, each $s_a : V \rightarrow \mathbb{R}$ continuous and real linear.
<code>frameDet B</code>	$\det M_F$, where $(M_F)_{ij} = \ell_i(e_j)$ and e_j are the standard coordinate vectors.
In the source	Mathematical meaning
<code>coeff a i; replacementDet B i (sat a)</code>	The real number $(c_a)_i$ and $\det M_F[i \leftarrow s_a]$, replacing row i by $(s_a(e_j))_j$.
<code>: ℝ := weight * frameDet B + ∑ a : J, ∑ i : Fin n, ...</code>	The output is the real number $w \det M_F + \sum_{a \in J} \sum_i (c_a)_i \det M_F[i \leftarrow s_a]$. Both sums range over their full finite index sets.

There is no model M in this signature. In particular, no contraction, sign of weight, or invertibility is an input condition. `coeff a i` is two successive function application, not a product.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–13:

```
private abbrev Coord (n : ℕ) := Fin n → ℝ
```

Exact source lines 15–18:

```
private abbrev DualFrame (n : ℕ) := DeterminantFrame.Frame n (Coord n)
private abbrev dualFrameMatrix {n : ℕ} := DeterminantFrame.frameMatrix (Pi.basisFun ℝ (Fin n))
private abbrev frameDet {n : ℕ} := DeterminantFrame.frameDeterminant (Pi.basisFun ℝ (Fin n))
private abbrev replacementDet {n : ℕ} := DeterminantFrame.replacementDeterminant (Pi.basisFun ℝ (Fin n))
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: MathlibAnnex.Satellite.satellitePolynomial

Accepted content SHA-256: 7f1e367358fd0f9445e65155eb71ac09f762d2092a022b0976220b0009be4cef

Accepted source guide SHA-256: 87ecf606c5b515afdb6d64d8d5ebc76efe3e2c7843f82e8fadc33b2b69e129b0

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73